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ASPECT-BASED SENTIMENT ANALYSIS ON HOTEL REVIEWS

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1. Problem Introduction

The demand for building automated functions for the application of the Ministry of Culture, Sports and Tourism arises from the broader trend of digital transformation and automation in customer-feedback analysis. The problem addressed in this project is the prediction of service-related indicators based on textual reviews and user ratings, thereby supporting recommendations and operational improvement strategies.

Within the scope of this report, the team focuses on the problem of **Aspect-Based Sentiment Analysis** (*ABSA*) on hotel reviews.

2. Data Collection Objectives and Scope

The data collection stage aims to build a hotel-review dataset that is sufficiently large, reliable, and aligned with the intended deployment context. The selected source is the Agoda booking platform, focusing on hotels in Da Nang for a hypothetical stay period from **June 1, 2025** to **June 7, 2025**. The main tools used are Python and Selenium WebDriver for crawling, together with Pandas and Langdetect for the initial preprocessing steps.

Table 1: Data collection setup

Attribute	Description
Data source	Agoda
Study area	Da Nang City
Hypothetical stay period	June 1, 2025 – June 7, 2025
Data type	Hotel metadata and user reviews
Collection tools	Python, Selenium WebDriver
Preprocessing tools	Pandas, Langdetect
Target output	Vietnamese hotel-review dataset for ABSA

3. Data Collection Strategy

The data collection strategy is divided into two phases: collecting hotel metadata and collecting detailed reviews. This design ensures that the dataset contains both high-level contextual information about each hotel and detailed textual feedback for sentiment analysis.

3.1. Collection of hotel metadata

In the first phase, Selenium is used to handle Agoda's *lazy-loading* mechanism by combining page scrolling with appropriate waiting times so that all hotel cards are fully loaded before extraction. The data is collected page by page until the full candidate list is exhausted.

To ensure both representativeness and preliminary quality, the following initial filters are applied:

- Hotels missing important fields such as rating or review count are removed.
- Only hotels with a total number of reviews between **100** and **1000** are retained to avoid extremely sparse or overly diluted cases.

This phase yields **402 hotels** that satisfy the initial filtering criteria.

3.2. Collection of detailed reviews

From the filtered hotel list, the system visits each hotel URL to extract detailed review content. During this process, the team prioritizes Agoda's built-in translation feature in order to obtain Vietnamese text for international reviews, thereby improving language consistency in the collected data.

Hotels without reviews or hotels that fail during page loading are skipped to prevent interruptions in the large-scale crawling process. After completion, the collected dataset contains **392 hotels** with a total of **41,344 raw reviews**.

Table 2: Data collection results by phase

Item	Count
Hotels satisfying metadata criteria	402
Hotels with successfully collected reviews	392
Total raw reviews	41,344

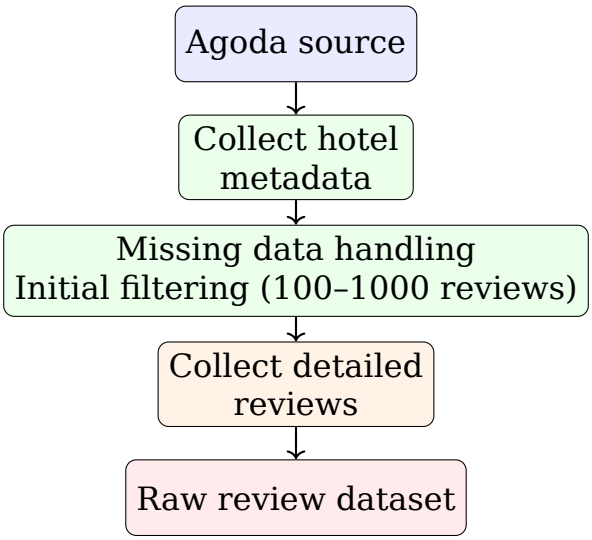

Figure 1: Data collection pipeline from Agoda

4. Data Preprocessing and Cleaning

After collection, the raw dataset is further cleaned to remove noise, improve language consistency, and prepare the data for downstream machine learning models.

4.1. Core cleaning steps

The main cleaning operations are summarized as follows:

- **Language detection:**
  - The `langdetect` library is used to retain only Vietnamese reviews.
  - **Objective:** Remove noise from English reviews or reviews written in other languages that are not relevant to Vietnamese NLP model training.
- **Deduplication:**
  - Duplicate records are removed based on `review_id`.
- **Empty-data filtering:**
  - Records with missing textual content (NaN or empty strings) or missing rating values are removed.
- **Heuristic noise filtering:**
  - Threshold-based rules are applied to remove overly short reviews, which typically provide little value for aspect-level analysis, as well as reviews with an excessively high ratio of special characters or digits relative to alphabetic characters.

After passing through the filtering stages above, the dataset shows a clear reduction in size:

- **Initial raw data:** 41,344 records.
- **Data after cleaning:** 37,327 records (a reduction of approximately 9.7%).

4.2. Five-layer processing pipeline

After the initial filtering stage, the data is passed through five additional processing layers.

4.2.1. Layer 1 – Basic text cleaning

The first layer removes direct sources of textual noise:

- Removing very short reviews (fewer than 5 words), duplicates (by content), and empty values.
- Using regular expressions to remove HTML, emoji, URLs, email addresses, phone numbers, and hashtags.
- Retaining only Vietnamese characters and basic digits while normalizing whitespace.

Example: “ks dep quá!!! xem tại http://abc.com → “ks dep quá

4.2.2. Layer 2 – Vietnamese typing and slang normalization

The second layer standardizes Vietnamese writing conventions:

- Unifying Unicode representations into a single standard form.
- Normalizing wrongly placed tone marks.
- Mapping common abbreviations and slang, for example `ks` to “khách sạn,” `nv` to “nhân viên,” and `ko` to “không.”

Example: “ks nay ko tot lam → “khách sạn này không tốt lắm

4.2.3. Layer 3 – AI-based spelling correction

To handle spelling and grammar errors at scale, the system integrates the `bmd1905/vietnamese-correction-v2` Transformer Seq2Seq model. Compared with manual rule writing, this approach is more automated and flexible for the diverse error patterns found in real reviews. Inference is optimized with batch processing, mixed precision, and fast decoding strategies suitable for tens of thousands of reviews. *Example:* “phòg sach se nhưng nhân vien chua nhiet tinh → “phòng sạch sẽ nhưng nhân viên chưa nhiệt tình; “hài hoà → “hài hòa

4.2.4. Layer 4 – Word segmentation for machine learning

Vietnamese contains many multi-syllabic words and compound expressions, so word segmentation is a necessary step before feeding the text into language models or classical feature-based pipelines. This layer improves

input consistency and reduces vocabulary fragmentation. *Example:* “nhân viên thân thiện → “nhân_viên thân_thiện

4.2.5. Layer 5 – Label and aspect normalization

In the final preprocessing stage, the label space is restructured to better support the ABSA task. The system groups 34 fine-grained aspect labels into 6 broader categories aligned with the hotel-review domain: **Cleanliness**, **Hotel**, **Rooms**, **Food & Drinks**, **Location**, and **Service**. At the same time, sentiment labels are reduced from 5 to 3 classes: **0 – Not mentioned**, **1 – Positive**, and **2 – Negative**. The grouping rule prioritizes negative sentiment in ambiguous cases so that important service issues are not overlooked. *Example:* `"Wifi": negative, "internet": negative` → `Service: 2`

Table 3: Summary of the multi-layer preprocessing pipeline

Layer	**Processing step**	Main objective
1	**Basic cleaning**	Remove noisy records, HTML, emoji, URLs, extra symbols, and very short texts
2	**Vietnamese typing normalization**	Standardize Unicode, tone marks, and slang dictionaries
3	**AI-based spelling correction**	Automatically fix spelling and grammar errors at scale
4	**Word segmentation**	Normalize model input and reduce vocabulary fragmentation
5	**Aspect-label normalization**	Group aspects and reduce sentiment labels to 3 classes

Table 4: Dataset size across preprocessing and labeling stages

Data milestone	Rows	Notes
Input after crawling	37,324	Input dataset after crawling and overall cleaning
After preprocessing	35,183	Reduced by 2,141 rows compared with the input
After labeling	35,183	No row-count change compared with the post-preprocessing dataset

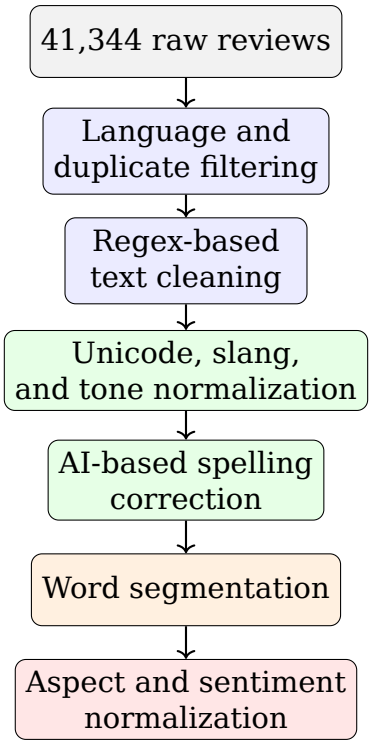

Figure 2: Preprocessing and data-cleaning pipeline

5. Analysis and Processing of the VLSP 2018 Hotel Training Data

The PhoBERT model used in the current stage is trained on the **VLSP 2018 Hotel** benchmark dataset. However, the original VLSP labels are highly fragmented, so before training, the team performs a major restructuring step to convert the dataset into a more suitable format for real-world Aspect-Based Sentiment Analysis.

5.1. Why the VLSP data needs restructuring

The original data is organized in the form "Entity–Attribute," producing approximately **34 aspect classes** such as `ROOMS#CLEANLINESS`, `HOTEL#DESIGN`, and `FOOD#QUALITY`. This design leads to an extremely sparse label matrix, because a real review rarely addresses all these aspects simultaneously. As a result, the model struggles to learn stable signals and its representational capacity becomes overly dispersed.

5.2. Grouping 34 aspect classes into 6 major aspects

To address this issue, the team groups the fine-grained labels into **6 broader aspects** that are better aligned with real hotel-review insights:

- **Cleanliness**: aggregates all cleanliness-related attributes across rooms, hotels, and facilities.
- **Hotel**: includes general, hotel-level attributes, including formerly ambiguous labels such as `FACILITIES`.
- **Rooms**: covers room-related attributes and room amenities while separating them from cleanliness to avoid overlap.
- **Food & Drinks**, **Location**, and **Service**: remain mostly intact because they already form distinct categories.

5.3. Reducing sentiment labels to 3 classes

In addition to reducing the number of aspects, the sentiment-label space is also simplified. The original dataset contains several sentiment states such as positive, negative, neutral, conflict, and not mentioned. To improve training consistency, the project reduces them to **3 final classes**:

- **0**: Not mentioned.
- **1**: Positive.
- **2**: Negative.

When multiple sub-columns are merged into a larger aspect, the system applies a *strict negative* rule: if positive and negative counts are tied, or if negative slightly dominates, the merged label is assigned as **negative**. This helps the model prioritize service-related issues rather than flattening important negative signals.

Table 5: VLSP dataset statistics after 3-class normalization

Split	Role	Not mentioned	Positive	Negative
Train	Main knowledge source for model training	8852	6948	2200
Dev	In-training evaluation and convergence monitoring	6625	3982	1393
Test	Independent final evaluation	1722	1377	501

Overall, the VLSP restructuring step transforms a highly fragmented label space into a more compact and informative representation for machine learning. This forms the foundation that enables PhoBERT to achieve better accuracy when used as a large-scale pseudo-labeler for real-world hotel reviews.

6. Analysis of the Labeling Model

The project uses **PhoBERT** as the core architecture for multi-aspect sentiment labeling. During development, the team experimented with three approaches before selecting the final configuration.

6.1. Three tested approaches

- **Approach 1 – Basic baseline**: retains the original aspect set and the full sentiment-label space. This approach performs poorly because the label space is too fragmented, especially with ambiguous boundaries between labels such as neutral and conflict.
- **Approach 2 – Domain-aware refinement**: replaces some less meaningful aspects with *Cleanliness* and starts using *Focal Loss*. Although it is more consistent with the hotel domain, performance remains limited because many noisy labels are still retained.
- **Approach 3 – Optimized 3-class setting**: reduces the entire sentiment space to three classes – *not mentioned*, *positive*, and *negative*. This final approach is selected because it allows the model to focus on the most meaningful sentiment signals.

The combination of a cleaned aspect set, a compact label space, and a more suitable loss function allows PhoBERT to learn minority cases more effectively, especially the negative class.

In the training stage, the model uses *Focal Loss* in the following form:

- Sample-wise cross-entropy: $CE_i = -\log(p_t)$
- Focal term: $FL_i = \alpha(1 - p_t)^\gamma CE_i$
- Mini-batch average loss:

$$L = \frac{1}{N} \sum_i FL_i$$

The values used in the implementation are $\alpha = 1.0$ and $\gamma = 2.0$.

6.2. Technical optimization

The technical setup is designed to improve both training stability and hardware efficiency:

- **Loss function**: *Focal Loss* is used instead of *Cross Entropy* to address label imbalance.
- **Memory optimization**: *Gradient Accumulation* and *Mixed Precision* are used to increase the effective batch size on GPU.
- **Training monitoring**: *Early Stopping* is applied based on F1-score and accuracy to prevent overfitting.

Table 6: PhoBERT performance by aspect

Aspect	Accuracy	Macro F1
Cleanliness	0.8183	0.8043
Food and drinks	0.9417	0.9143
Hotel	0.7733	0.7388
Location	0.9450	0.8423
Rooms	0.8367	0.8339
Service	0.9117	0.8415

The overall model performance reaches **Accuracy = 0.9141** and **Macro F1 = 0.8813**, indicating strong general effectiveness across the full aspect-labeling task.

The results show that the model performs especially well on aspects such as *Food & Drinks*, *Location*, and *Service*, while *Hotel* and *Cleanliness* remain more challenging because of broader semantics and higher overlap. Nevertheless, the overall performance is strong enough for PhoBERT to act as a high-confidence pseudo-labeler for the crawled dataset.

6.3. Fine-tuning Process and Training Hyperparameters

To automate the annotation process for the newly crawled dataset, the pre-trained **PhoBERT** model (`vinai/phobert-base`) was fine-tuned on the benchmark VLSP 2018 Hotel dataset. The fine-tuning process adapts the robust linguistic representations of PhoBERT to the specific nuances of Aspect-Based Sentiment Analysis (ABSA).

Fine-tuning Workflow:

- **Data Representation:** Input reviews are first pre-processed (including Unicode normalization and word segmentation using VnCoreNLP/BPE) and tokenized to match PhoBERT's vocabulary. The

resulting token sequences are fed into the transformer encoder.

- **Model Adaptation:** While the core transformer layers are initialized with pre-trained weights, a task-specific classification head is appended to the top layer. Instead of training from scratch, the model fine-tunes these weights to map the contextualized `[CLS]` representation to the appropriate aspect-sentiment pairs.
- **Inference and Pseudo-labeling:** Once the optimal weights are achieved, the model is deployed as an automated annotator. It processes the large-scale, unlabelled crawled data to confidently predict aspect and sentiment labels, generating a high-quality dataset for downstream applications.

Hyperparameter Configuration:
Careful tuning of hyperparameters was crucial to ensure stable convergence and prevent overfitting. The final model was trained with the following settings:

- **Pre-trained Base:** `vinai/phobert-base`
- **Maximum Epochs:** 15
- **Batch Size:** 8 (chosen to maintain stable gradient updates within hardware constraints).
- **Peak Learning Rate:** 2×10^{-5} (implemented with a scheduler for smooth weight updates).
- **Evaluation Strategy:** The model was evaluated on the development set, and checkpoints were saved every 500 steps.
- **Early Stopping:** Configured with a patience of 3 evaluation steps. The system monitored the Macro F1-score to halt training automatically if no improvement was observed, effectively preventing the model from memorizing the training data.

Training Convergence:
According to the training logs, the model demonstrated efficient convergence. The optimal weights were captured at **Global Step 5625** (exactly at the end of Epoch 5.0). At this checkpoint, the model achieved its peak validation metrics (Accuracy = 0.9141, Macro F1 = 0.8813). This specific checkpoint was restored and utilized for all subsequent baseline evaluations and the final massive pseudo-labeling task.

7. Post-Label Data Processing

After PhoBERT assigns labels to the crawled dataset, the team performs a post-processing pipeline to produce a cleaner, more stable, and more auditable training dataset. Rather than simply discarding low-confidence samples, the pipeline is designed to both filter data for training and retain difficult cases for evaluation and later human labeling.

7.1. Objectives of the post-processing stage

This stage has three main goals:

1. Select the optimal confidence threshold on a validation set with ground truth.
2. Remove low-confidence pseudo-labels to reduce noise in the training data.
3. Build a *relabel pool* of approximately 3,000 difficult samples for subsequent *human labeling*.

From a technical perspective, the post-processing pipeline consists of three consecutive steps: **threshold calibration**, **confidence filtering**, and **rule-based noise filtering combined with relabel-pool creation**. This separation makes pseudo-label quality easier to evaluate and simplifies artifact tracking for each stage.

7.2. Confidence definition used in the pipeline

Each labeled sample comes with six probabilities corresponding to the six aspects. From these probabilities, the team defines two aggregate confidence measures:

- `conf_mean`: the mean of the six probabilities, representing the overall confidence of the sample.
- `conf_min`: the minimum of the six probabilities, representing the weakest aspect-level confidence.

Using both measures is important because a sample may have a relatively high average confidence while still containing one very weak aspect prediction. In multi-aspect learning, even one incorrect local label can degrade the training signal.

7.3. Confidence-threshold calibration

The team scans thresholds 0.70, 0.75, 0.80, and 0.85 on the VLSP *dev* split. For each threshold, only samples satisfying `conf_mean` $\geq$ `threshold` are retained, and predictions are compared against ground truth to compute *macro F1* and *micro F1*.

Table 7: Confidence-threshold comparison on the dev set

Threshold	Kept rows	Kept ratio	F1-macro	F1-micro
0.70	1957	0.9785	0.8451	0.8838
0.75	1866	0.9330	0.8522	0.8905
0.80	1558	0.7790	0.8635	0.9037
0.85	896	0.4480	0.8872	0.9295

From these results, the team selects **threshold_best = 0.85** because it achieves the highest macro F1 and is consistent with the goal of prioritizing data cleanliness over maximum sample retention.

7.4. Confidence filtering

When applied to the crawled dataset, a sample is retained only if it satisfies both conditions:

- `conf_mean` ≥ 0.85
- `conf_min` ≥ 0.5

The second condition acts as a safety floor for the weakest aspect. It prevents cases where the average confidence is deceptively high while at least one aspect is still close to random.

Table 8: Confidence-filtering results on the crawled dataset

Metric	Value
Before filtering	35,183
After filtering	17,295
Removed	17,888
Retention	49.16%
Mean conf_mean	0.8416 $\rightarrow$ 0.8869
Mean conf_min	0.6854 $\rightarrow$ 0.7814

7.5. Rule-based noise filtering

On top of confidence filtering, the team applies several manual rules to remove samples with a high probability of being noisy or uninformative:

- `rule_full_none`: all aspects are labeled 0.
- `rule_all_same_pos`: all aspects are labeled 1.
- `rule_all_same_neg`: all aspects are labeled 2.
- `rule_text_too_short`: the review is too short, with `token` < 4 or `char` < 15.

Table 9: Noise-filtering results after confidence filtering

Metric	Value
Step-3 input	17,295
Step-3 keep	16,309
Step-3 removed	986
Step-3 retention	94.30%
rule_full_none	731
rule_all_same_pos	102
rule_all_same_neg	0
rule_text_too_short	158
Mean conf_mean	0.886866 $\rightarrow$ 0.887279
Mean conf_min	0.781388 $\rightarrow$ 0.782985

These results show that Step 3 removes only a small fraction of data, but in a highly selective way, making the final retained set more stable without substantially changing the overall confidence distribution.

7.6. Intermediate step before human labeling: the 3k relabel pool

Instead of fully discarding the removed samples, the team keeps them to build a *relabel pool* of about **3,000 samples**, focused on the difficult region of the data distribution. The sampling strategy is divided into three groups:

- **Hard sample** (about 70%): samples with `conf_mean` in the difficult range 0.75–0.80 or with `conf_min` < 0.6.
- **Medium confidence** (about 20%): samples with `conf_mean` in the

range 0.80–0.90.

- **Random sample** (about 10%): randomly selected from both borderline removed cases and part of the retained set.

Table 10: Actual composition of the 3k relabel pool

Component	Count
Total samples	3000
Hard	2100
Medium	566
Random	274
Fallback	60

The `fallback` component appears when some buckets do not contain enough unique samples or have too much duplication. In those cases, the script automatically fills the gap from the remaining pool while still preserving a strong emphasis on difficult samples.

7.7. Role of the relabel pool

Keeping and relabeling difficult samples helps prevent the model from becoming over-specialized on easy, overly clean, and overly confident cases. In particular, this step provides three major benefits:

1. It reduces overfitting to easy *high-confidence* samples.
2. It injects human feedback into the *decision boundary*, where pseudo-labels are most error-prone.
3. It reduces systematic pseudo-label bias in ambiguous or sentiment-conflict cases.

7.8. Evaluation of post-processing quality

After the filtering and relabel-pool construction stages, the team evaluates data quality through three main questions. This part no longer remains at the planning stage; instead, it directly uses the most recent audit results to answer each question.

In the current setting, the crawled data no longer has an original pseudo-label snapshot by version, so *before/after* comparison is not possible. There-

fore, the evaluation is conducted in **single-system audit** mode, meaning that the current system is evaluated directly against human labels.

7.8.1. Question 1: Is the difficult-sample selection strategy working correctly?

The *sample error rate* by group is:

- fallback: 0.8500
- hard: 0.8257
- medium: 0.7633
- random: 0.7044

The most important pairwise tests are also statistically significant:

- hard vs. random: $p = 2.0369 \times 10^{-6}$
- hard vs. medium: $p = 9.0628 \times 10^{-4}$
- fallback vs. random: $p = 3.2191 \times 10^{-2}$

These results indicate that the current sampling strategy is indeed **capturing the difficult region**. The *hard* and *fallback* groups are genuinely harder than the *random* group, so Q1 is considered successful.

7.8.2. Question 2: What is the absolute quality of the current system?

Since there is no earlier pseudo-label baseline for comparison, Q2 is interpreted as an absolute quality measurement of the current system against human labels. The overall results are:

- macro F1: 0.7299
- micro F1: 0.7564

The weakest aspect is **hotel**, with a macro F1 of only 0.5723. This suggests that the current system is already usable within the pipeline, but its quality is still uneven across aspects.

7.8.3. Question 3: Is the current system biased toward the none class?

The *none over-prediction ratio* by aspect is:

- service: 1.4977
- cleanliness: 1.4388
- location: 1.2315
- rooms: 1.1229
- food and drinks: 0.9634
- hotel: 0.5468

Values above 1 indicate that the model still over-predicts *none* for several aspects, most notably **service**, **cleanliness**, and **location**. Therefore, Q3 shows that the *none-bias* issue has not yet been fully resolved.

7.8.4. Artifacts for verification and traceability

Each stage of the post-processing pipeline produces its own artifacts so that the workflow remains auditable and reproducible. In the filtering stage, files are organized by Step 1, Step 2, and Step 3, including threshold-scan results, scored outputs, retained outputs, removed outputs, and summary files.

After the evaluation stage, the system also produces dedicated audit artifacts for Q1–Q3, such as group-level error summaries, pairwise statistical tests, absolute-quality reports by aspect, and reports on *none*-class bias. Preserving all these artifacts is essential because it turns post-processing from a simple cleaning stage into a reproducible quality-control framework.

8. Outputs of the Data Collection and Processing Stages

From the above pipeline, the team obtains a structured input dataset with sufficient quality for automatic labeling by PhoBERT. The main value of this stage lies in three key outcomes:

1. The data collection process is clearly aligned with both the target problem and the practical deployment context.
2. The multi-layer cleaning pipeline significantly reduces noise in Vietnamese user reviews.
3. The normalization of aspects and sentiment labels provides a strong foundation for training and evaluating ABSA models in the next stages.

Overall, the data collection and processing stage serves as the foundation of the entire aspect-based sentiment analysis pipeline. The quality of these steps directly determines the reliability of the labeling model and the usefulness of the final training dataset in subsequent stages.

9. Exploratory Data Analysis

9.1. Data Description

The dataset is divided into three main splits:

- **Training set:** 13,224 samples.
- **Validation set:** 3,778 samples.
- **Test set:** 1,889 samples.

The dataset contains the following columns:

- **Review:** Review text written in Vietnamese.
- **Rating:** Overall rating score ranging from 1 to 10.
- **Aspect labels:** Six aspect-specific label columns: vệ sinh_label, đồ ăn thức uống_label, khách sạn_label, vị trí_label, phòng ốc_label, and dịch vụ_label.
  - **Label convention:** 0 = not mentioned/neutral, 1 = positive, and 2 = negative.

9.2. Data Quality Check

9.2.1. Missing Values and Duplicates

No missing values are found in any critical fields across all three splits (training, validation, and test). In addition, no duplicated records are detected in the dataset.

9.2.2. Label Inconsistency

A cross-check is conducted between the overall rating scores and the negative labels:

- **Rating $=$ 10 with at least one negative label:** 281 samples (2.13%).
- **Rating $\geq$ 9 with at least two negative labels:** 59 samples (0.45%).
- **Rating $\leq$ 4 with at least two positive labels:** 21 samples (0.16%).
- **Conclusion:** The inconsistency rate is very low. Manual inspection suggests that these cases mostly reflect mixed-sentiment reviews, in which users praise one aspect while criticizing another, rather than annotation errors.

9.3. Statistical Analysis

9.3.1. Rating Distribution Analysis

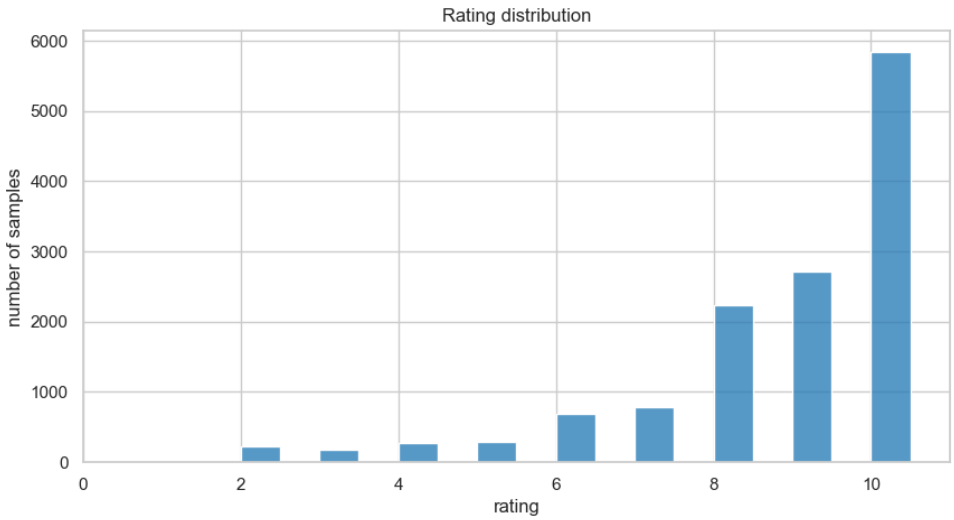

Figure 3: Distribution of overall rating scores in the dataset

- **Mean:** 8.61.
- **Median:** 9.0.
- **Distribution pattern:** The rating distribution is left-skewed, indicating that most customers tend to give positive ratings in the 8–10 range. Low ratings (2–5) account for only a small proportion of the dataset.

9.4. Text Statistics

Review length (token count):

- **Min:** 3 tokens.
- **Max:** 205 tokens.
- **Mean:** 31 tokens.

Keyword frequency (overall and by aspect)

WordCloud from training reviews
phòng
khách_sạn
nhân_viên
tuyệt_vời
thân_thiện
chúng_tôi
sạch_sẽ
biển
vị_trí
thoải_mái
nhiệt_tình
tiện_nghi
đẹp
đà_nẵng
giá_cả
dịch_vụ
giường
nghỉ
bãi
rộng_rãi
có_thể
hợp_lý
phải_chăng
lựa_chọn
đầy_đủ
thuận_tiện
chu_đáo
chuyên_nghiệp
khu_vực
ngủ
bữa
phút
tuyệt
hàng
chất_lượng
khuyến
giúp_đỡ
giá_trị
dễ_thương
vấn_đề
nhìn_chung
thuận_lợi
cà_phê
hiện_đại
nhược_điểm
dễ_chịu
vì_vậy
thật_sự
lễ_tân
tầng
món
view
ban_công
đối_ngũ
thăm_chí
xung_quanh
phù_hợp
thực_sự
hoàn_toàn
mặc_dù
khe
buffet
hành_lý
gọi
dễ_dàng
thiết_kế
vui_vẻ
vệ_sinh
mong_đợi
tiện_lợi
hoàn_hảo
thoáng
nghỉ_ngơi
rộng
phục_vụ
đánh_giá
nằm
chỗ
hơi
không_gian
hỗ_trợ
bắt_tiền
chợ
kém
một_chút
cảm_giác
tất_cả
cảm_ơn
đi_lại
hồ
chủ
cửa
đồ_ăn
tiền
rẻ
đường
thăm_quan
lưu_trú
nhà_vệ_sinh
muộn
ngoại_trừ
lịch_sử
phòng_không
thang_máy
giới_thiệu
hoạt_động
tắm
khăn
căn_hộ
du_lịch
đầu_tiên
bắn
ok
thời_gian
tiếng
mùi
sân_bay
hải_lòng
chắc_chắn
bể_bơi
vật_chất
thám_chí
view
cũ
ánh
quầy
giá_trị
khuyến
chất_lượng
món
tầng
đối_ngũ
ban_công
khe
nóng
nhà_nghỉ
thuê
bao
hoàn_toàn
mặc_dù
phù_hợp
lắm
gọi
tế
hành_lý
buffet

9.5. Aspect Label Distribution

Table 11: Percentage distribution of sentiment labels across aspects

Aspect	**0**	**1**	**2**
dịch vụ_label	52.155	43.724	4.121
khách sạn_label	23.503	68.618	7.880
phòng ốc_label	57.887	29.598	12.515
vệ sinh_label	70.395	23.858	5.747
vị trí_label	63.793	34.082	2.125
đồ ăn thức uống_label	82.880	13.090	4.031

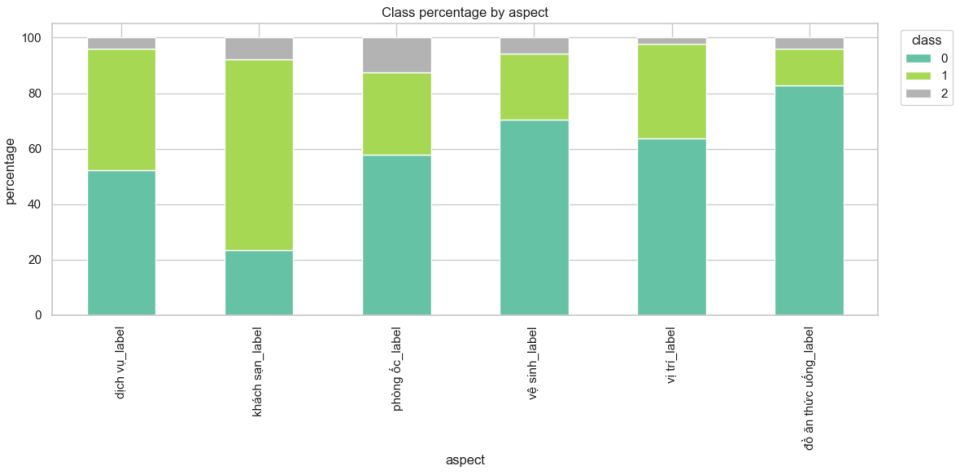

Figure 6: Sentiment-label proportions across aspects

- **Mention frequency:** The most frequently discussed aspects are Hotel (76.5%), Service (47.8%), and Room (42.1%), whereas Food & Beverage is the least mentioned aspect (17.1%).
- **Top strengths:** User satisfaction is particularly high for the overall Hotel experience (68.62% positive), followed by Service (43.72%) and Location (34.08%).
- **Major pain points:** Room emerges as the main weakness, with the highest negative rate overall (12.5%).
- **Data sparsity in class 0:** The dominance of class 0 across multiple aspects indicates a large number of generic reviews that do not mention specific aspect categories.

10. Models, Approaches, and Evaluation

10.1. Modeling Objectives

The model is designed to solve the multi-label Aspect-Based Sentiment Analysis (ABSA) task in the hotel domain. Specifically, the model needs to simultaneously predict the sentiment scores for six independent aspects: *Hygiene, Food and Beverage, Hotel, Location, Room,* and *Service*.

To comprehensively address this problem and evaluate different methodologies, the research team implemented three distinct approaches:

- **Approach 1:** A traditional machine learning method utilizing Support Vector Machine (SVM) combined with TF-IDF for text feature extraction.
- **Approach 2:** A parameter-efficient deep learning method using ViDeBERTa combined with Houlsby Adapters, Multi-Pooling, and Multi-Sample Dropout (MSD).
- **Approach 3:** A hybrid sequence-learning method using PhoBERT with BiGRU and masked mean pooling, formulated as a multi-head classification problem with six aspect-specific heads.

10.2. Approach 1: Linear Support Vector Machine

This method utilizes a traditional machine learning pipeline, leveraging the linear kernel of the Linear Support Vector Machine (LinearSVC) combined with TF-IDF vectorization for multi-aspect classification. To ensure high computational efficiency and interpretability on local hardware while effectively handling the multi-label nature of hotel reviews, the model incorporates the following techniques:

- **Architecture**
  - **Feature Extraction:** The processed Vietnamese text is converted into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency). This method captures the importance of specific sentiment-bearing words while downscaling common stop words.
  - **Classifier:** The core model is a Linear Support Vector Machine (LinearSVM). To handle the multi-aspect nature of the problem, a One-Vs-Rest (OvR) strategy is employed, where a separate binary SVM classifier is trained for each of the six hotel aspects.
- **Training Setup**
  - **Hyperparameters:** The SVM uses a linear kernel to maintain high-dimensional efficiency with text data. Regularization ($C = 1.0$) is applied to balance the margin maximization and classification error.
  - **Multi-output Strategy:** To address the multi-label nature of Aspect-Based Sentiment Analysis, where one review can contain multiple aspects, the system utilizes a MultiOutputClassifier wrapper. This wrapper encapsulates the base estimator and fits one independent classifier for each of the six target aspects.
  - **One-Vs-Rest (OvR) Mechanism:** Within each aspect's classifier, because each aspect has three possible sentiment classes, the model employs a One-Vs-Rest strategy. For each aspect, the system trains three separate binary classifiers (e.g., Positive vs. All others). During inference, the class with the highest confidence score from its respective binary units is selected as the final sentiment label for that specific aspect.
  - **Implementation:** The pipeline is implemented using the scikit-learn library, with TfidfVectorizer configured for n-grams (1, 2) to capture local word context like "không tốt" or "rất sạch".

Table 12: Hyperparameter configuration of the TF-IDF + Linear SVM

Hyperparameter	**Value**
Feature Extractor	TF-IDF Vectorizer
N-gram Range	(1, 2)
Normalization	L2 norm
Max Features	50000
Multi-label Strategy	MultiOutputClassifier
Base Estimator	Linear Support Vector Machine (LinearSVC)
Loss Function	Squared Hinge
Penalty	L2
Regularization (C)	1.0
Multi-class Strategy	One-Vs-Rest (OvR)
Max Iterations	1000
Random Seed	42

10.3. Approach 2: ViDeBERTa + Adapter + Multi-Pooling + MSD

This approach applies fine-tuning to the Vietnamese language model ViDeBERTa-base (Fsoft-AIC/videberta-base) for multi-aspect classification. To improve computational efficiency while preserving the ability to learn complex contextual patterns in hotel reviews, the model incorporates the following techniques:

- **Efficient Fine-Tuning (PEFT):** Instead of fine-tuning the entire architecture, the model employs Houlsby bottleneck adapters with a bottleneck dimension of 64, inserted before the LayerNorm layers in the attention and feed-forward blocks. The backbone is frozen except for the adapter layers, the LayerNorm parameters, and the last two encoder blocks. This design reduces the number of trainable parameters to 16.6 million, or 8.9% of the total parameters, thereby substantially lowering memory costs.
- **Multi-Pooling Strategy:** To extract richer semantic features, the model does not rely solely on the traditional [CLS] token. Instead, the hidden states are processed through a multi-pooling mechanism that combines [CLS], mean pooling, and max pooling, together with a dedicated mask that excludes the [CLS] and [SEP] tokens.
- **Multi-Sample Dropout (MSD):** To improve generalization and reduce overfitting, the multi-pooled representation is passed through five dropout branches with different rates (0.05, 0.10, 0.15, 0.20, 0.25). The final loss is computed as the average of the losses from these branches.
- **Training Procedure:** The model uses weighted cross-entropy loss with label smoothing (0.1) to address class imbalance across the Positive, Negative, and Neutral/Not Mentioned labels. Training is performed with the AdamW optimizer under a discriminative fine-tuning strategy, using a learning rate of 5e-5 for the last two encoder blocks and 2e-4 for the adapters and classification heads. The model is trained with a maximum sequence length of 256 and a batch size of 32. Mixed precision training (AMP) is enabled on two Tesla T4 GPUs using DataParallel, and training runs for up to 20 epochs with early stopping patience set to 3.

Table 13: Hyperparameter configuration of the fine-tuned ViDeBERTa model

Hyperparameter	**Value**
Backbone	ViDeBERTa-base (DeBERTa-v2)
Adapter bottleneck dim	64
Multi-pooling	CLS + Mean + Max
MSD dropout rates	{0.05, 0.10, 0.15, 0.20, 0.25}
Optimizer	AdamW
Learning rate (adapters/heads)	2e-4
Learning rate (encoder layers)	5e-5
Batch size	32
Max sequence length	256 tokens
Max epochs	20
Warmup radio	0.1
Label smoothing	0.1
Early stopping patience	3 epochs
Gradient clipping	max norm = 1.0
Random seed	42

10.4. Approach 3: PhoBERT + BiGRU + Mean Pooling + Multi-Head Classification

This method combines the contextual language representation power of PhoBERT (originally pre-trained as `vinai/phobert-base-v2`) with the sequential learning capabilities of a BiGRU. The model architecture (`PhoBERT_MultiHeadGRU`) consists of the following components:

- **Feature Extractor (Backbone):** The model uses a pre-trained Vietnamese language encoder, PhoBERT, loaded from `vinai/phobert-base-v2`. All parameters in the PhoBERT encoder are fine-tuned end-to-end during training, rather than freezing the backbone. The input review text is tokenized and truncated or padded to a maximum length of `MAX_LEN` = 128 tokens.
- **Sequential Context Layer (BiGRU):** The contextual token representations produced by PhoBERT (`last_hidden_state`) are passed into a bidirectional GRU with hidden size 256. Because the GRU is bidirectional, it learns dependencies from both left-to-right and right-to-left contexts, which helps refine the sentence representation before classification.
- **Pooling Strategy (Masked Mean Pooling):** After the BiGRU layer, the output token representations are aggregated into a single fixed-length vector by masked mean pooling. This pooling strategy uses the attention mask to exclude padding tokens, so only real tokens contribute to the final sentence representation. This is more reliable than plain mean pooling because it prevents padding from affecting the representation.
- **Classification Heads:** The pooled sentence vector is fed into six independent classification heads, one for each aspect label. Each head outputs logits over three classes {0, 1, 2}, and the model is trained using `CrossEntropyLoss`. During inference, the predicted class for each aspect is obtained by taking the `argmax` over the corresponding head output.

Table 14: Hyperparameter configuration of the fine-tuned PhoBERT-BiGRU multi-head classifier

Hyperparameter	**Value**
Backbone	PhoBERT-base-v2
Sequential layer	BiGRU
Pooling strategy	Masked mean pooling
Classification heads	6 aspect-specific heads $\times$ 3 classes
Loss function	CrossEntropyLoss
Optimizer	AdamW
Learning rate	2×10^{-5}
Batch size	16
Max sequence length	128 tokens
Max epochs	5
GRU hidden dimension	256
Gradient clipping	max norm = 1.0
Random seed	42
Checkpoint selection metric	Validation mean aspect macro-F1
Saved checkpoint	`best_phobert_gru_ce.pth`

10.5. Evaluation Metrics

To comprehensively evaluate model performance, especially under the severe class imbalance commonly found in real-world review datasets, the following metrics are used:

- **Macro F1-Score (Primary Metric):** This metric is computed by averaging the F1-scores of the three labels (0, 1, and 2) for each aspect independently, regardless of their frequencies in the dataset. It provides a fairer assessment of model performance by preventing the model from appearing strong simply because it favors the majority class (for example, when most reviews are labeled as neutral). The scores are also aggregated into the mean macro F1 across all six aspects to represent overall performance.
- **Accuracy:** This metric measures the percentage of correctly predicted samples out of the total number of evaluated samples for each aspect.
- **Computational Performance Metrics:** These include training time per epoch and inference time per sample, which are used to assess the practical feasibility of real-world deployment.

10.6. Evaluation Results

10.6.1. Approach 1: Linear Support Vector Machine

The evaluation results for the Linear SVM baseline model, using TF-IDF features and a Multi-Output strategy. The model was evaluated on the same test set of 1,889 samples to ensure a fair comparison with the deep learning approach.

Table 15: Aspect-level performance of the Linear Support Vector Machine model on the test set

Aspect	Macro F1	Accuracy
Hygiene	0.8853	0.9449
Food	0.8223	0.9539
Hotel	0.7035	0.7941
Location	0.8178	0.9423
Room	0.8631	0.8957
Service	0.8654	0.9412

- **Overall Performance:** The model achieves a average macro-F1 score of 0.8263 and a average Accuracy of 0.912 across all six aspects.
- **Computational Cost:** The training process for all six aspects is highly efficient, taking approximately 1 to 3 minutes to complete on a local CPU.

10.6.2. Approach 2: ViDeBERTa + Adapter + Multi-Pooling + MSD

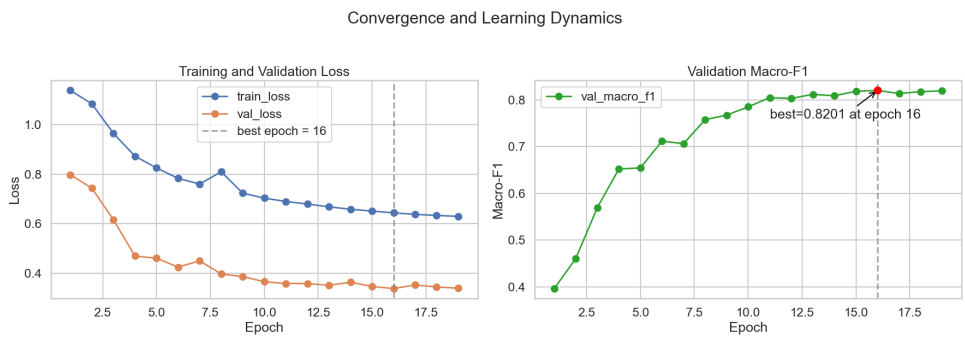

Figure 7: Training/validation loss and validation macro-F1 for the fine-tuned ViDeBERTa model

Note: During training, the model is optimized using class-weighted cross-entropy loss with label smoothing to address class imbalance and reduce overfitting. In contrast, during validation, standard cross-entropy loss is used to measure performance more objectively.

The following table presents the evaluation results of the fine-tuned ViDeBERTa model on the test set (1889 samples), after training stopped early at epoch 19, with epoch 16 identified as the best-performing epoch.

Table 16: Aspect-level performance of the fine-tuned ViDeBERTa model on the test set

Aspect	**Macro F1**	**Accuracy**
Hygiene	0.8500	0.9269
Food	0.8595	0.9576
Hotel	0.6924	0.7983
Location	0.7889	0.9344
Room	0.8578	0.8925
Service	0.8420	0.9328

- **Overall Performance:** The model achieves a average macro-F1 score of 0.8151 and a average Accuracy of 0.907 across all six aspects.
- **Computational Cost:** Training takes approximately 3.1 minutes per epoch (equivalent to about 188 seconds) on a system with two Tesla T4 GPUs. During inference, the model requires approximately 20.10 ms per sample. Including validation after each epoch, the total wall-clock training time was about 74 min 26 s

10.6.3. Approach 3: PhoBERT + BiGRU + Mean Pooling + Multi-Head Classification

Table 17 reports the evaluation results of the fine-tuned PhoBERT + BiGRU 6-head model on the held-out test set containing 1,889 samples. The model was trained for 5 epochs, and the best checkpoint was selected at epoch 4 based on the validation mean-aspect F1 score.

Training Dynamics: During training, the loss decreased steadily from 0.3774 in epoch 1 to 0.0889 in epoch 5. Validation performance improved up to epoch 4, where the model achieved its best validation result with a mean-aspect F1 of 0.8933 and a validation loss of 0.1763. After epoch 4, validation performance slightly declined in epoch 5, suggesting that the

Table 17: Test performance of the PhoBERT + BiGRU 6-head model

Aspect	**Accuracy**	**Macro-F1**
Hygiene	0.9502	0.9155
Food & Beverage	0.9698	0.8985
Hotel	0.8857	0.8213
Location	0.9555	0.8667
Room	0.9481	0.9346
Service	0.9603	0.9092

checkpoint from epoch 4 was the most effective one.

Training Time: The training time per epoch was approximately 3.9–4.3 minutes, with epoch times of 3 min 54 s, 4 min 17 s, 4 min 11 s, 4 min 15 s, and 4 min 07 s, respectively. The total training time for 5 epochs was approximately 20 min 44 s. Including validation after each epoch, the total wall-clock training time was about 22 min 36 s.

Overall Performance: The model achieved an overall accuracy of 0.9449, a test macro-F1 score of 0.9101, a mean-aspect macro-F1 score of 0.8910, and an exact-match ratio of 0.7454 on the test set. These results indicate that the model performs well at the individual label level, while full-label exact prediction across all six aspects remains more challenging.

Additional Observations:

- The *Room* aspect achieved the highest performance on the test set, with a macro-F1 score of 0.9346.
- The *Hotel* aspect was the most difficult to predict, with a macro-F1 score of 0.8213, indicating greater ambiguity compared to the other aspects.
- The exact-match ratio of 0.7454 means that approximately 74.54% of the test samples were predicted correctly across all six aspects simultaneously.
- Inference on the test set took approximately 11 seconds for 1,889 samples, corresponding to about 5.8 ms per sample.

10.7. Discussion

10.7.1. Approach 1: Linear Support Vector Machine

The Linear SVM-based approach with TF-IDF features and a multi-output strategy performs effectively as a baseline on the ABSA task in the hotel domain, as reflected by a average macro-F1 score of 0.8263.

- **Strengths:**
  - **Computational Efficiency:** LinearSVM is extremely fast to train and requires minimal memory compared to Transformer-based models.
  - **Interpretability:** The model weights for each aspect can be easily inspected to identify which keywords (e.g., ”wifi”, ”bẩn”, ”nhiệt tình”) contribute most to a specific sentiment prediction
  - **No Overfitting on Small Data:** Due to its high bias and simple structure, the model is robust against noise in smaller datasets or aspects with very few samples.
- **Weeknesses:**
  - **Lack of Semantic Context:** TF-IDF treats words as independent units, failing to understand complex Vietnamese grammar, sarcasm, or long-range dependencies that models like ViDeBERTa or PhoBERT can capture.
  - **Limited Generalization:** The model relies heavily on exact word matches, making it less effective at handling spelling variations or slang that were not explicitly covered in the preprocessing/dictionary normalization stage.

10.7.2. Approach 2: ViDeBERTa + Adapter + Multi-Pooling + MSD

The ViDeBERTa-based approach with adapters and MSD performs effectively on the ABSA task in the tourism domain, as reflected by a average macro-F1 score of 0.8151.

- **Strengths:**
  - **Feature extraction capability:** Clearly defined physical aspects such as Food, Room, and Hygiene achieve very high F1 scores (> 0.85). The multi-pooling mechanism helps the model capture short sentiment-bearing spans that may be distributed across the review and could be overlooked when relying solely on the [CLS]

token.

  - **Deployment feasibility:** Owing to the adapter mechanism, the number of trainable parameters is reduced to fewer than 17 million, enabling efficient mixed-precision training in only about 3 minutes per epoch. The inference speed of 20.10 ms per sample is also highly practical, supporting near-real-time responsiveness in real-world applications.
  - **Robustness:** Multi-Sample Dropout (MSD) helps the model mitigate overfitting effectively. This is reflected in the steady decrease and stable convergence of the validation loss before early stopping is triggered.
- **Weaknesses and observations:**
  - **The anomalous behavior of the "Hotel" aspect:** The Hotel aspect records the lowest macro F1 (0.6924) and the lowest accuracy (0.7983).
  - **Subtle class imbalance:** The Location aspect shows very high accuracy (0.9344) but a lower macro F1 of 0.7889. This suggests a heavily skewed label distribution: reviewers usually either praise the location or do not mention it, while negative comments about location are relatively rare. Although class weights are used, the scarcity of negative samples for the Location aspect remains an empirical challenge.

10.7.3. Approach 3: PhoBERT + BiGRU + Mean Pooling + Multi-Head Classification

The PhoBERT-based model with a BiGRU layer and six aspect-specific classification heads performs strongly on the tourism-domain ABSA task, achieving a mean macro-F1 score of 0.8910 on the test set. The best checkpoint is obtained at epoch 4, with a validation mean-aspect macro-F1 of 0.8933.

- **Strengths:**
  - **Strong contextual modeling:** PhoBERT provides rich Vietnamese contextual representations, while the BiGRU layer helps preserve token order and capture sentiment transitions that may occur away from the `[CLS]` token. This combination leads to strong overall performance, including 0.9449 accuracy and 0.7454 exact-match on the test set.
  - **Aspect-aware prediction:** The six parallel heads allow the

model to learn separate decision boundaries for *vệ sinh*, *đồ ăn thức uống*, *khách sạn*, *vị trí*, *phòng ốc*, and *dịch vụ*. In practice, this design yields particularly high scores for *phòng ốc* (macro-F1 0.9346), *vệ sinh* (0.9155), and *dịch vụ* (0.9092).
  - **Stable optimization:** The model is straightforward to train with AdamW, cross-entropy loss, and gradient clipping. The relatively low validation loss at the best checkpoint (0.1763) suggests that the training process converges cleanly before overfitting becomes severe.
- **Weaknesses and observations:**
  - **Semantic ambiguity in the *khách sạn* aspect:** This aspect obtains the lowest macro-F1 (0.8213). A plausible explanation is that general comments about the hotel are often mixed with opinions about room quality, service, or location, making the boundary between aspects less clear.
  - **Location remains relatively uneven:** Although *vị trí* achieves high accuracy (0.9555), its macro-F1 is lower at 0.8667. This pattern typically indicates a skewed label distribution, where positive or neutral mentions are much more frequent than negative ones.
  - **Full-instance prediction is still challenging:** The exact-match ratio is 0.7454, which means that predicting all six aspects correctly in a single review remains difficult even when the per-aspect scores are strong.
  - **Heavier than parameter-efficient alternatives:** Because the entire PhoBERT backbone is fine-tuned, this approach is more computationally demanding than adapter-based methods and may be less lightweight for deployment.

10.8. Overall Discussion and Insights

The results provide several important insights into the trade-offs between performance, efficiency, and modeling capability.

Performance vs. Complexity Trade-off A key observation is that increased model complexity does not necessarily guarantee better performance. Despite being the simplest approach, the Linear SVM baseline achieves a macro-F1 score of 0.8263, outperforming the more sophisticated ViDeBERTa-based model (0.8151). This suggests that, for structured domains such as hotel reviews—where sentiment is often expressed through

explicit keywords—lexical features like TF-IDF remain highly competitive.

In contrast, the PhoBERT + BiGRU model clearly outperforms both alternatives, achieving the highest macro-F1 score of 0.8910 and a strong exact-match ratio of 0.7454. This indicates that when sufficient modeling capacity is combined with an appropriate architectural design, deep learning approaches can significantly improve fine-grained sentiment classification.

The Role of Contextual Representation The comparative results highlight the importance of contextual understanding in ABSA. The SVM model relies solely on surface-level lexical features, making it effective for explicit sentiment cues but incapable of capturing deeper semantics. The ViDeBERTa model introduces contextual embeddings; however, its reliance on parameter-efficient fine-tuning (PEFT) limits its adaptability to domain-specific patterns.

In contrast, the PhoBERT-based model, which fully fine-tunes the backbone and incorporates a BiGRU layer, achieves a better balance between contextual representation and sequential modeling. This allows the model to capture long-range dependencies and subtle sentiment transitions more effectively.

These findings suggest that contextual representation alone is insufficient; the way contextual features are further processed (e.g., through sequence models such as BiGRU or specialized classification heads) plays a critical role in downstream performance.

Aspect-Level Challenges and Data Characteristics Across all approaches, certain aspects consistently exhibit lower performance, particularly the *Hotel* aspect. This reflects inherent challenges related to data characteristics rather than model limitations. The *Hotel* aspect is semantically broad and often overlaps with other aspects (e.g., room, service), leading to label ambiguity.

Similarly, the *Location* aspect demonstrates class imbalance, where positive or neutral labels dominate. This results in high accuracy but comparatively lower macro-F1 scores. These patterns suggest that data-centric improvements—such as refined annotation guidelines, better aspect disambiguation, or data augmentation—are crucial for further performance gains.

Multi-Aspect Learning and Prediction Difficulty Another important finding is the gap between per-aspect performance and full-instance prediction. Although the PhoBERT + BiGRU model achieves high macro-F1 scores at the individual aspect level, the exact-match ratio remains at 0.7454. This indicates that predicting all six aspects correctly within a single review remains a challenging task.

This reflects the inherent difficulty of multi-output learning, where an error in any single aspect leads to failure in exact-match evaluation. Future work may explore advanced strategies such as modeling dependencies between aspects, hierarchical or joint learning frameworks, and calibration techniques to improve prediction consistency.

Efficiency and Practical Deployment From a practical perspective, each approach occupies a different position along the efficiency–performance spectrum. The Linear SVM model is highly efficient and interpretable, making it suitable for lightweight applications or rapid prototyping. The ViDeBERTa model with adapters provides a balance between efficiency and contextual modeling, offering reasonable performance with reduced computational cost.

In contrast, the PhoBERT + BiGRU model delivers the best predictive performance but at a higher computational cost, making it more appropriate for production environments where accuracy is the primary concern. These findings highlight that model selection should be guided by application constraints rather than performance alone.

Key Takeaways

- Traditional methods such as SVM remain strong baselines for structured sentiment analysis tasks.
- Parameter-efficient fine-tuning does not always yield optimal performance when model adaptability is constrained.
- Combining contextual embeddings with sequential modeling (PhoBERT + BiGRU) is highly effective for ABSA.
- Data characteristics, including ambiguity and class imbalance, significantly impact model performance.
- Multi-aspect prediction remains inherently challenging despite strong per-aspect results.

11. Application

This section is reserved for the application stage, where the project output is connected to practical usage scenarios.

11.1. Application Objectives

This subsection should explain the purpose of the application layer and the user or business problem it addresses.

11.2. System Workflow

This subsection can describe how the model output is integrated into an application workflow, dashboard, or decision-support pipeline.

11.3. Potential Use Cases

This subsection can present practical use cases, such as service monitoring, review summarization, sentiment-based recommendations, or operational alerts.

11.4. Limitations in Deployment

This subsection should discuss practical limitations, such as latency, domain shift, model drift, or interpretability concerns.

11.5. Future Application Directions

This subsection can outline possible improvements or extensions for real-world deployment.

12. References

References

[1] https://arxiv.org/pdf/1706.03762

[2] https://arxiv.org/pdf/2006.03654

[3] https://arxiv.org/pdf/2111.09543

[4] https://arxiv.org/pdf/2301.10439

[5] https://huggingface.co/Fsoft-AIC/videberta-base

[6] https://arxiv.org/pdf/1902.00751

[7] https://arxiv.org/pdf/1905.09788